

Mass as Information Resistance

An Information-Geometric Framework for the Derivation of the Einstein Equations and their Quantum Implications

Document I in the FisherGeometrics framework

Based on an exploratory session, 28 February 2026

Working document — speculative theoretical research

1. Starting point: Mass as information resistance

The departure point of this research is a reinterpretation of mass (or more broadly: the stress-energy tensor $T_{\mu\nu}$) in the Einstein field equations. Rather than treating mass as a fundamental property of matter, we propose that mass is an *emergent phenomenon*: it reflects the resistance of the information medium against information flux.

This idea connects to several existing research directions:

- **Erik Verlinde** (2010) proposes that gravity is an entropic effect.
- **Ted Jacobson** (1995) derived the Einstein equations from thermodynamic relations ($\delta Q = T dS$) on causal horizons.
- The **holographic principle** (Bekenstein, 't Hooft, Susskind) links information to surfaces in spacetime.

What distinguishes this framework from earlier approaches is the explicit construction around three pillars: Fisher information as a tensor, information density as a local field, and entropy as the driving force.

2. The three fundamental ingredients

2.1 Fisher information as a tensor

The first crucial insight is that information must be a tensor in order to couple to the stress-energy tensor $T_{\mu\nu}$. A scalar measure such as Shannon entropy can never replace the full structure of $T_{\mu\nu}$, which has 10 independent components in 4D spacetime.

The Fisher information matrix is ideally suited as an information tensor:

$$\mathcal{F}_{\mu\nu} = \mathbb{E} \left[\frac{\partial \ln p}{\partial \theta^\mu} \cdot \frac{\partial \ln p}{\partial \theta^\nu} \right] \quad (1)$$

The Fisher information matrix is a Riemannian metric on a statistical manifold (Rao, 1945; Amari, 1985). This yields:

- covariantly transforming tensor structure,
- an intrinsic notion of distance between states (Cramér-Rao bound), and
- a natural curvature on the information manifold.

The Fisher metric generates its own curvature, which we can identify with spacetime curvature.

2.2 Information density as a local field

Two regions with the same total information but different densities must curve spacetime differently — just as the same mass concentrated in a sphere of 1 km versus 1 light-year produces a completely different metric. The information density ρ_I plays the role of a local “compaction degree”: the more densely packed the information, the greater the curvature.

This promotes the Bekenstein bound ($S \leq 2\pi RE/\hbar c$) from an inequality to a dynamical equation.

2.3 Entropy as the driving force

The third pillar is entropy S as the potential that drives information flux. The de Bruijn identity connects entropy and Fisher information:

$$\frac{dS}{dt} = \frac{1}{2} \mathcal{F} \quad (2)$$

The time derivative of entropy is Fisher information. This completes the triangle:

Concept	Role	Analogy
Fisher information $\mathcal{F}_{\mu\nu}$	Metric structure, resistance	Impedance
Information density ρ_I	Local concentration	Charge density
Entropy S	Driving potential	Potential difference

The constitutive relation becomes an **information Ohm's law**:

$$\text{Flux} = \frac{\text{Entropic gradient}}{\text{Fisher resistance}} \quad (3)$$

Or in tensor form:

$$J^\mu = \mathcal{F}^{\mu\nu} \cdot \nabla_\nu S \cdot \rho_I \quad (4)$$

3. Local time as information transport duration

A particularly powerful insight in this framework is that proper time is not a fundamental quantity, but **the time information needs to traverse one coordinate step through the information medium**.

Define a local information speed:

$$v_I(r) = \frac{|\nabla S|}{\mathcal{F}_{rr} \cdot \rho_I} \quad (5)$$

- Far from a mass: low Fisher resistance, low information density $\rightarrow$ information flows fast $\rightarrow$ clocks tick normally.
- Near the horizon: $\mathcal{F}_{rr} \rightarrow \infty$ and $\rho_I \rightarrow \infty$, so $v_I \rightarrow 0 \rightarrow$ information takes an infinitely long time to make one step $\rightarrow$ **time stops**.

This is gravitational time dilation — not as an effect on clocks, but as a fundamental property of the information medium.

3.1 Implications

The speed of light as information speed

c is not an arbitrary constant but a consequence of the vacuum information structure:

$$c = v_I^{\max} = \frac{|\nabla S|_{\text{vac}}}{\mathcal{F}_{\text{vac}} \cdot \rho_{I,\text{vac}}} \quad (6)$$

In vacuum without mass, the Fisher resistance is minimal and uniform $\rightarrow c$ is constant and isotropic. Mass locally increases the resistance $\rightarrow v_I < c \rightarrow$ time dilation.

Causal structure = information structure

The light cone is the surface where $v_I = c$. The horizon of a black hole is where $v_I = 0$ in the outward direction. Causality is not an axiom but a consequence of information dynamics.

Time is not fundamental

In standard GR, the metric (including g_{tt}) is a fundamental field. In this framework, g_{tt} is derived from the information structure. Time emerges from the rate at which information flows through the medium. No information medium $\rightarrow$ no time.

4. Derivation of the Schwarzschild metric

4.1 Setting and target metric

Static, spherically symmetric, mass M at the origin, outside the source ($T_{\mu\nu} = 0$). The metric to be reproduced:

$$ds^2 = -f(r) c^2 dt^2 + f(r)^{-1} dr^2 + r^2 d\Omega^2 \quad (7)$$

with $f(r) = 1 - r_s/r$ and $r_s = 2GM/c^2$.

4.2 Constitutive relations

Relation 1 — Spatial metric from information resistance:

$$g_{rr} = \frac{\mathcal{F}_{rr}}{\rho_I} \quad (8)$$

Relation 2 — Information flow speed:

$$v_I = \frac{dS/dr}{\mathcal{F}_{rr} \cdot \rho_I} \quad (9)$$

Relation 3 — Time metric from information speed:

$$-g_{tt} = v_I^2 \cdot g_{rr} \quad (10)$$

4.3 The information fields

Fisher information — A static observer at radius r sits in a thermal bath with Tolman temperature $T(r) = T_H/\sqrt{f(r)}$:

$$\mathcal{F}_{rr}(r) = \mathcal{F}_0 \cdot f(r)^{-2} \quad (11)$$

Information density — scales with the Tolman temperature:

$$\rho_I(r) = \rho_0 \cdot f(r)^{-1} \quad (12)$$

Entropy gradient — the entropy density of the thermal field scales as T^3 :

$$\frac{dS}{dr}(r) = S'_0 \cdot f(r)^{-2} \quad (13)$$

4.4 The derivation

Radial component g_{rr} :

$$g_{rr} = \frac{\mathcal{F}_{rr}}{\rho_I} = \frac{\mathcal{F}_0 \cdot f^{-2}}{\rho_0 \cdot f^{-1}} = \frac{\mathcal{F}_0}{\rho_0} \cdot f(r)^{-1} \quad (14)$$

With normalisation $\mathcal{F}_0/\rho_0 = 1$:

$$\boxed{g_{rr} = \frac{1}{1 - r_s/r}} \quad \checkmark \quad (15)$$

Information speed:

$$v_I = \frac{S'_0 \cdot f^{-2}}{\mathcal{F}_0 \cdot f^{-2} \cdot \rho_0 \cdot f^{-1}} = \frac{S'_0}{\mathcal{F}_0 \rho_0} \cdot f(r) \quad (16)$$

With $S'_0/(\mathcal{F}_0 \cdot \rho_0) = c$:

$$\boxed{v_I(r) = c \left(1 - \frac{r_s}{r}\right)} \quad \checkmark \quad (17)$$

Time component g_{tt} :

$$-g_{tt} = v_I^2 \cdot g_{rr} = c^2 f^2 \cdot f^{-1} = c^2 f \quad (18)$$

$$\boxed{g_{tt} = -c^2 \left(1 - \frac{r_s}{r}\right)} \quad \checkmark \quad (19)$$

The full metric:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad \checkmark \quad (20)$$

This is the Schwarzschild solution, derived from information geometry.

5. Normalisation conditions and their meaning

Condition	Equation	Meaning
Flat space	$\mathcal{F}_0 = \rho_0$	In vacuum, resistance = density (equilibrium)
Speed of light	$S'_0 = c \cdot \mathcal{F}_0 \cdot \rho_0$	c is the information flux speed in vacuum

The second normalisation is particularly deep: the speed of light is not a free constant, but is determined by the ratio of the vacuum entropy gradient to the vacuum information resistance times the information density.

6. Behaviour at the horizon

Quantity	Limit $r \rightarrow r_s$	Interpretation
$\mathcal{F}_{rr}$	$\rightarrow \infty$	Infinite information resistance
ρ_I	$\rightarrow \infty$	Maximal information packing (Bekenstein bound)
dS/dr	$\rightarrow \infty$	Maximal entropic driving force
v_I	$\rightarrow 0$	Information stops
g_{tt}	$\rightarrow 0$	Time stops

The black hole is the point where information resistance exactly overcomes the entropic driving force and brings information flow to a halt.

Inside the horizon ($r < r_s$), $f(r)$ becomes negative: $v_I < 0$ (information can only flow inward) and g_{rr} and g_{tt} swap signs (r becomes timelike, t becomes spacelike). This follows automatically from the sign behaviour of the information fields.

7. The de Bruijn equilibrium condition

The de Bruijn identity $dS/dt = \frac{1}{2}\mathcal{F}$ tells us that in a diffusive system the rate of entropy growth equals the Fisher information. In the static case ($\partial_t = 0$), the system is in informational equilibrium:

$$\nabla_\mu J_I^\mu = 0 \quad (21)$$

This is the information analogue of hydrostatic equilibrium: the entropy gradient pushing information outward is exactly balanced by the Fisher resistance opposing information flux.

Gravitational waves are then propagating disturbances of this equilibrium — ripples in $\mathcal{F}_{\mu\nu}$, ρ_I and S that propagate at speed v_I .

8. Hawking radiation in the information framework

A crucial aspect of the framework is that Hawking radiation does not need to be added separately — it is already implicitly present. The Tolman temperature $T(r) =$

$T_H/\sqrt{f(r)}$, based on the Hawking temperature:

$$T_H = \frac{\hbar c^3}{8\pi G M k_B} \quad (22)$$

8.1 Hawking radiation as information leakage

At the horizon the Fisher resistance is classically infinite and the information speed zero. But quantum-mechanically the resistance is not exactly infinite — the quantum Fisher information is always finite. There is a small but nonzero information leak through the barrier.

That leak is Hawking radiation: information tunnelling through the Fisher resistance barrier, driven by the entropy gradient. The de Bruijn identity guarantees that as long as the Fisher information is not exactly zero, the system produces entropy.

8.2 Evaporation and the information paradox

As the black hole radiates, M decreases, r_s decreases, and the Fisher resistance at the horizon decreases. The leak grows — exactly the known result that $T_H \propto 1/M$.

In the information framework **there is no paradox**: information is not lost. It is stored in the Fisher resistance structure and leaks through gradually. Unitarity is guaranteed by the conservation law $\nabla_\mu J_I^\mu = 0$.

8.3 The Page curve

The Page curve describes how the entropy of the Hawking radiation first increases and then decreases. In this framework it follows from the dynamics of ρ_I . Initially the information density at the horizon is so high that the emitted radiation is barely correlated (high entropy). Halfway through the evaporation (the “Page time”), the information density begins to decrease and the correlations in the radiation become visible — the entropy falls.

9. Polarity and unification with electromagnetism

9.1 Polarity reversal at the horizon

Inside the black hole $f(r)$ becomes negative: v_I reverses and g_{rr} and g_{tt} swap roles. This is literally a polarity reversal of the information medium.

9.2 Driving force versus resistance as polarity

The entropy gradient pushes information outward (∇S points away from the mass), while the Fisher resistance opposes this. These two opposing “polarities” create the equilibrium that yields the static metric. Disturb that equilibrium and you get gravitational waves.

9.3 Electric charge as information polarity

If the information medium has a polarity, the question arises whether electric charge is also an information polarity phenomenon:

- **Gravity** is the scalar aspect of the information medium — the total resistance and density, regardless of orientation. That is why gravity is always attractive: more information = more resistance = more curvature. There is no “negative mass” because information density is always positive.
- **Electromagnetism** is the vectorial aspect — the orientation of the information flux. Two equal orientations repel; two opposite orientations attract.

9.4 Symmetric and antisymmetric

The electromagnetic field tensor $F_{\mu\nu}$ is antisymmetric, while the Fisher metric is symmetric:

Aspect of the information field	Correspondence
Symmetric part	Gravity (via Einstein)
Antisymmetric part	Electromagnetism (via Maxwell)

This is the splitting that Kaluza and Klein attempted in the 1920s using a fifth dimension — but here it emerges naturally from two aspects of information: *how much* (density, scalar) versus *in which direction* (orientation, vectorial).

10. The information core of black holes

10.1 Maximal information density

Maximal information density is reached when the Fisher resistance hits its quantum limit. The Planck length $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m defines the scale. The Bekenstein-Hawking entropy states: 1 bit of information fits in 4 Planck areas.

The maximal information density is not volumetric — it is **holographic**.

10.2 The core

Information cannot be compressed to a point — the Fisher resistance diverges before $r = 0$ is reached. There is a minimal radius, of order the Planck length, at which the information density reaches its maximum.

	Atom	Black hole
Core	Nucleus (quarks, gluons)	Planck sphere (maximal ρ_I)
Shell	Electron cloud	Horizon (maximal resistance)
Binding	Electromagnetic force	Fisher resistance
Radiation	Photon emission	Hawking radiation
Stability	Quantum pressure	Information degeneracy pressure

10.3 Information degeneracy pressure

The fact that you cannot pack more than 1 bit into a Planck volume yields an opposing pressure: an **information degeneracy pressure**, analogous to electron degeneracy pressure in white dwarfs or neutron degeneracy pressure in neutron stars.

Degeneracy pressure	Object
Electron degeneracy	White dwarf
Neutron degeneracy	Neutron star
Information degeneracy	Planck core in a black hole

10.4 The information trajectory through a black hole

Phase	Location	Description
1 — Approach	$r > r_s$	Fisher resistance grows as $f(r)^{-2}$, information density as $f(r)^{-1}$. Information speed decreases.
2 — Horizon crossing	$r = r_s$	Quantum-mechanically finite but extreme resistance. Phase transition: polarity reverses.
3 — Interior	$r < r_s$	r becomes timelike. Information moves only toward $r = 0$.
4 — The core	$r \sim \ell_P$	Maximally compressed information in a Planck sphere. Not singular but maximally ordered.
5 — Return	Hawking radiation	Quantum tunnelling through the Fisher resistance barrier. Information leaks slowly outward.

10.5 Excited states

If black holes have a core like atoms, they also have excited states. Transitions between quantum states of the core produce discrete frequencies in the Hawking radiation — an “atomic spectrum of spacetime.”

A black hole is an information atom — the largest atom in the universe, with the smallest core. The Planck mass $m_P \approx 2.18 \times 10^{-8}$ kg is the point where the information medium says: “no further.”

11. Quantum scale: Schrödinger as a 1D projection

11.1 The problem with Schrödinger

The Schrödinger equation is first order in time, second order in space — time and space are treated fundamentally unequally. In the information framework that is a problem: time is not a fundamental parameter but the duration of information transport.

Schrödinger describes information flux in one direction — forward in time. But the information medium has a full tensor structure with 10 components. Schrödinger effectively uses only one. This directly explains why Schrödinger is incompatible with relativity.

11.2 The information wave equation

The quantum Fisher information of a state ψ is identical to the Fubini-Study metric on the projective Hilbert space — fully covariant. The evolution is described as information flux through the Fisher medium:

$$\mathcal{F}^{\mu\nu} \nabla_\mu \nabla_\nu \psi = \rho_I \cdot \frac{\partial S}{\partial \psi} \quad (23)$$

Equation	Information structure	Limitation
Schrödinger	Only $\mathcal{F}_{tt}$	Non-relativistic, 1D flux
Klein-Gordon	$\mathcal{F}_{\mu\nu} = \eta_{\mu\nu}$ (flat)	No gravity
Dirac	$\mathcal{F}_{\mu\nu} = \eta_{\mu\nu} + \text{spin}$	No gravity
Einstein	Classical $\mathcal{F}_{\mu\nu}$	No quantum
Information eq.	Full quantum $\mathcal{F}_{\mu\nu}$	None (?)

11.3 Implications for quantum phenomena

- **Entanglement** is not “spooky action at a distance” but a correlation in the Fisher information tensor that is local in the full tensor structure — it only appears non-local in the 1D projection.
- **The measurement problem** — the collapse of the wave function — is the result of a sudden change in the local information density ρ_I at a measurement. Not a mysterious process but information dynamics.
- All “strange” aspects of quantum mechanics — superposition, entanglement, the

measurement problem — are artefacts of the fact that Schrödinger is a projection of a higher-dimensional information process.

12. Core formulation

The spacetime geometry is the effective description of an information medium in which entropy is the driving force, Fisher information is the resistance, and information density is the local capacity. Mass, energy, and momentum are emergent phenomena from this information dynamics.

The four principles of the framework:

1. Spacetime curvature = information curvature (Fisher metric)
2. Mass = information resistance $\times$ information density
3. Entropy = driving force behind information flux
4. Time = duration of information transport through the medium

Three fields $(\mathcal{F}_{\mu\nu}, \rho_I, S)$, two constitutive relations, one metric. The Schwarzschild solution follows exactly.

13. Predictions and open questions

13.1 What this framework predicts

Regularisation of the singularity. The singularity at $r = 0$ would be regularised if the quantum Fisher information is used instead of the classical one — QFI is always finite for quantum states.

Cosmological constant. Λ would follow from the background entropy production rate of the vacuum. De Sitter space as thermal equilibrium of the information field.

Dark matter. Possibly a region with anomalous information density without corresponding baryonic matter — a purely information structure in the Fisher metric that generates curvature.

13.2 Open questions

- **Microscopic origin of $\mathcal{F}_{\mu\nu}$:** For $T_{\mu\nu}$ we know it comes from varying the matter action with respect to the metric. What is the “information action” from which $\mathcal{F}_{\mu\nu}$ follows?
- **Full tensor formulation:** The constitutive relation $g_{\mu\nu} = \Phi(\mathcal{F}_{\mu\nu}, \rho_I, \nabla S)$ must be formulated fully tensorially, not component by component.
- **Probability distribution family:** What is the fundamental family $p(x|\theta)$ whose Fisher metric yields the spacetime metric?
- **Gravitational waves:** Can propagating disturbances in the information field reproduce the full tensor structure of gravitational waves?

Note: these questions are answered in subsequent documents (II–XLII). The scalar-to-tensor upgrade and the manifold $K = \mathbb{CP}^2 \times S^3 \times S^1$ resolve the first three; gravitational waves follow from the information action (document IV).

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Document I of the FisherGeometrics framework (28 February 2026).

Working document. Continued in documents II–LVIII.